

DEFINITE INTEGRATION

Definition:

$\int_a^b f(x) dx = [F(x)]_a^b = F(b) - F(a)$ is called the definite integral of $f(x)$ between the limits a and b .

where $\frac{d}{dx}(F(x)) = f(x)$

Note : The word limit here is quite different as used in differential calculus.

Important Points:

(I) If $\int_a^b f(x) dx = 0$, then the equation $f(x) = 0$ has atleast one root in (a, b) provided f is continuous in (a, b) .

Note that the converse is not true.

Illustration :

$\int_0^1 e^x (ax^2 + bx + c) dx = 0 \Rightarrow e^x (ax^2 + bx + c) = 0$ has at least one root in $(0, 1)$
 $\Rightarrow ax^2 + bx + c = 0$ has at least one root in $(0, 1)$ [e^x is always positive]

(II) $\lim_{n \rightarrow \infty} \left(\int_a^b f_n(x) dx \right) = \int_a^b \left(\lim_{n \rightarrow \infty} f_n(x) \right) dx$

Illustration :

If $\lim_{n \rightarrow \infty} \int_{-\sqrt[n]{a}}^{\sqrt[n]{a}} \left(1 - \frac{t^3}{n} \right) t^2 dt = \frac{2\sqrt{2}}{3}$ ($n \in \mathbb{N}$), then the value of find 'a'.

Sol. L.H.S. = $\int_{-a^{1/3}}^{a^{1/3}} \lim_{n \rightarrow \infty} \left(1 - \frac{t^3}{n} \right) t^2 dt = \int_{-a^{1/3}}^{a^{1/3}} e^{-t^3} t^2 dt$

$$= \left[-\frac{1}{3} e^{-t^3} \right]_{-a^{1/3}}^{a^{1/3}} = \frac{1}{3} [e^a - e^{-a}] \Rightarrow e^a - e^{-a} = 2\sqrt{2} \text{ or } a = \ln(\sqrt{2} + \sqrt{3})$$

(III) $\int_a^b f(x) \cdot d(g(x)) = \int_{g^{-1}(a)}^{g^{-1}(b)} f(x) \cdot g'(x) dx$.

(IV) If $f(x)$ is continuous in (a, b) , Then $\int_a^b \frac{d}{dx}(f(x)) = [f(x)]_a^b$ and if $f(x)$ is discontinuous in (a, b) at

$$x = c \in (a, b), \text{ then } \int_a^b \frac{d}{dx}(f(x)) = [f(x)]_a^{c^-} + [f(x)]_{c^+}^b$$

Illustration :

$$\int_{-1}^1 \left(\frac{d}{dx} \left(\cot^{-1} \frac{1}{x} \right) \right) dx = \left[\cot^{-1} \frac{1}{x} \right]_{-1}^{0^-} + \left[\cot^{-1} \frac{1}{x} \right]_{0^+}^1 = \pi - \left(\frac{3\pi}{4} \right) + \frac{\pi}{4} = \frac{\pi}{2}$$

(V) If $g(x)$ is the inverse of $f(x)$ and $f(x)$ has domain $x \in [a, b]$ where $f(a) = c$ and $f(b) = d$ then the value

$$\text{of } \int_a^b f(x) dx + \int_c^d g(y) dy = (bd - ac)$$

Illustration :

$$\text{Evaluate : } \int_0^1 e^{\sqrt{e^x}} dx + 2 \int_e^{e^{\sqrt{e}}} \ln(\ln x) dx$$

Sol. Consider $f: [0, 1] \rightarrow [e, e^{\sqrt{e}}]$, $f(x) = e^{\sqrt{e^x}}$ then $f^{-1}(x) = 2 \ln(\ln x)$

$$I = \int_0^1 e^{\sqrt{e^x}} dx + 2 \int_e^{e^{\sqrt{e}}} \ln(\ln x) dx \quad \text{hence } I = 1 \cdot e^{\frac{1}{\sqrt{e}}} - 0 \cdot e = e^{\sqrt{e}}$$

Evaluating definite integrals by finding antiderivatives :

Illustration :

$$\text{Evaluate : } \int_3^8 \frac{\sin \sqrt{x+1}}{\sqrt{x+1}} dx$$

$$\text{Sol. } \int \frac{\sin \sqrt{x+1}}{\sqrt{x+1}} dx = -2 \cos \sqrt{x+1}$$

$$\Rightarrow \int_3^8 \frac{\sin \sqrt{x+1}}{\sqrt{x+1}} dx = \left[-2 \cos \sqrt{x+1} \right]_3^8 = 2 (\cos 2 - \cos 3)$$

Illustration :

$$\text{Evaluate : } \int_0^{\pi/4} \cos 2x \sqrt{4 - \sin 2x} dx$$

$$\text{Sol. } \int \cos 2x \sqrt{4 - \sin 2x} dx, \text{ Put } 4 - \sin 2x = t \Rightarrow -2 \cos 2x dx = dt$$

$$\text{Integral becomes } \int -\frac{1}{2} \sqrt{t} dt = -\frac{1}{3} (t)^{3/2} = -\frac{1}{3} (4 - \sin 2x)^{3/2}$$

$$\Rightarrow \int_0^{\pi/4} \cos 2x \sqrt{4 - \sin 2x} dx = \left[-\frac{1}{3} (4 - \sin 2x)^{3/2} \right]_0^{\pi/4} = \frac{8 - 3\sqrt{3}}{3}$$

Illustration :

Evaluate : $\int_0^1 x \ln(1+2x) dx = \frac{3 \ln 3}{8}$

Sol. $\int_0^1 x \ln(1+2x) dx = \ln(1+2x) \cdot \frac{x^2}{2} - \int \frac{2}{1+x} \cdot \frac{x^2}{2} dx = \frac{x^2}{2} \ln(1+2x) - \int \left(\frac{x}{2} - \frac{1}{4} \right) + \frac{1}{4} \left(\frac{1}{1+2x} \right) dx$
 $= \frac{x^2}{2} \ln(1+2x) - \frac{x^2}{4} + \frac{1}{4}x - \frac{1}{8} \ln(1+2x) \Rightarrow \int_0^1 x \ln(1+2x) dx = \frac{3}{8} \ln 3$

Illustration :

The value of the integral $\int_0^{2008} \left(3x^2 - 8028x + (2007)^2 + \frac{1}{2008} \right) dx$ equals

(A) $(2008)^2$

(B) $(2009)^2$

(C) 2009

(D) 1

Sol. $\int \left(3x^2 - 8028x + (2007)^2 + \frac{1}{2008} \right) dx = x^3 - 4014x^2 + (2007)^2 x + \frac{x}{2008}$
 $\Rightarrow \text{value of integral} = \left[x^3 - 4014x^2 + (2007)^2 x + \frac{x}{2008} \right]_0^{2008} = 2009$

Illustration :

Evaluate : $\int_2^4 \frac{\sqrt{x^2-4}}{x^4} dx$

Sol. $\int \frac{\sqrt{x^2-4}}{x^2} dx = \int \frac{\sqrt{1-4/x^2}}{x^3} dx$ Put $1 - \frac{4}{x^2} = t \Rightarrow 8x^{-3} dx = dt$ integral becomes
 $\int \frac{1}{8} t^{1/2} dt = \frac{1}{12} t^{3/2} = \frac{1}{12} \left(1 - \frac{4}{x^2} \right)^{3/2} \Rightarrow \int_2^4 \frac{\sqrt{x^2-4}}{x^4} dx = \left[\frac{1}{12} \left(1 - \frac{4}{x^2} \right)^{3/2} \right]_2^4 = \frac{\sqrt{3}}{32}$

Illustration :

Evaluate : $\int_0^{\pi/4} x \sin^2 x dx$

Sol. Let $I = \int_0^{\pi/4} x \sin^2 x dx = \int_0^{\pi/4} \frac{x}{2} (1 - \cos 2x) dx$

Integrating by parts, we have

$$I = \left[\frac{x}{2} \left(x - \frac{\sin 2x}{2} \right) \right]_0^{\pi/4} - \int_0^{\pi/4} \frac{1}{2} \left(x - \frac{\sin 2x}{2} \right) dx = \frac{\pi}{8} \left(\frac{\pi}{4} - \frac{1}{2} \right) - \left[\frac{x^2}{4} + \frac{\cos 2x}{8} \right]_0^{\pi/4}$$

$$= \left(\frac{\pi^2}{32} - \frac{\pi}{16} \right) - \left(\frac{\pi^2}{64} - \frac{1}{8} \right) = \frac{\pi^2 + 8 - 4\pi}{64}$$

Illustration :

Find the value of $\int_0^{1/2} \frac{x \cos^{-1} x}{\sqrt{1-x^2}} dx$.

Sol. Let $I = \int_0^{1/2} \frac{x \cos^{-1} x}{\sqrt{1-x^2}} dx$

Let us put $x = \cos t$, $dx = -\sin t dt$. Also, when $x = 0$, then $t = \frac{\pi}{2}$ and when $x = \frac{1}{2}$, then $t = \frac{\pi}{3}$.

Thus, we have

$$\begin{aligned} I &= \int_{\pi/2}^{\pi/3} \frac{t \cos t}{\sin t} (-\sin t dt) - \int_{\pi/2}^{\pi/3} t \cos t dt \\ &= [-t \sin t]_{\pi/2}^{\pi/3} + \int_{\pi/2}^{\pi/3} \sin t dt = \frac{\pi}{2} - \frac{\pi}{3} \cdot \frac{\sqrt{3}}{2} - [\cos t]_{\pi/2}^{\pi/3} = \frac{\pi(\sqrt{3}-1)}{2\sqrt{3}} - \frac{1}{2}. \end{aligned}$$

Practice Problem

Q.1 Let $\int_0^1 \frac{dx}{\sqrt{16+9x^2}} + \int_0^2 \frac{dx}{\sqrt{9+4x^2}} = \ln a$. Find a .

Q.2 Evaluate: $\int_0^{\ln 2} x e^{-x} dx$

Q.3 Evaluate: $\int_0^{\pi/4} \frac{\sin^2 x \cdot \cos^2 x}{(\sin^3 x + \cos^3 x)^2} dx$

Q.4 Evaluate: $\int_0^{\pi/2} \frac{dx}{1 + \cos \theta \cdot \cos x}$ $\theta \in (0, \pi)$

Q.5 Suppose that the function f, g, f' and g' are continuous over $[0, 1]$, $g(x) \neq 0$ for $x \in [0, 1]$, $f(0) = 0$, $g(0) = \pi$, $f(1) = \frac{2009}{2}$ and $g(1) = 1$. Find the value of the definite integral,

$$\int_0^1 \frac{f(x) \cdot g'(x) \{g^2(x) - 1\} + f'(x) \cdot g(x) \{g^2(x) + 1\}}{g^2(x)} dx.$$

Answer key

Q.1 $(2^{1/3} \cdot 3^{1/2})$ Q.2 $1 - \frac{2}{e}$ Q.3 $\frac{1}{6}$ Q.4 $\left(\frac{\theta}{\sin \theta}\right)$

Q.5 2009

PROPERTIES OF DEFINITE INTEGRAL :

(A) PROPERTIES :

P - 1 : $\int_a^b f(x) dx = \int_a^b f(t) dt$; **P - 2 :** $\int_a^b f(x) dx = - \int_b^a f(x) dx$

P - 3 : $\int_a^b f(x) = \int_a^c f(x) dx + \int_c^b f(x) dx$ provided f has a piece wise continuity
or when f is not uniformly defined in (a, b)
Integral is broken at points of discontinuity or at the points where definition of ' f ' changes.

Illustration :

Evaluate : $\int_0^{\pi} \sqrt{\frac{1+\cos 2x}{2}} dx$.

Sol. $\int_0^{\pi} |\cos x| dx = \int_0^{\pi/2} \cos x dx + \int_{\pi/2}^{\pi} -\cos x dx = [\sin x]_0^{\pi/2} + [-\sin x]_{\pi/2}^{\pi} = 2$

Illustration :

Evaluate : $\int_0^3 |5x - 9| dx$.

Sol. $\int_0^3 |5x - 9| dx = \int_0^{9/5} (9 - 5x) dx + \int_{9/5}^3 (5x - 9) dx = \left[9x - \frac{5}{2}x^2 \right]_0^{9/5} + \left[\frac{5x^2}{2} - 9x \right]_{9/5}^3 = \frac{15}{2}$

Illustration :

Evaluate : $\int_{-1}^3 \left[x + \frac{1}{2} \right] dx$

Sol. $I = \int_{-1}^{-1/2} \left[x + \frac{1}{2} \right] dx + \int_{-1/2}^{1/2} \left[x + \frac{1}{2} \right] dx + \int_{1/2}^{3/2} \left[x + \frac{1}{2} \right] dx + \int_{3/2}^{5/2} \left[x + \frac{1}{2} \right] dx + \int_{5/2}^3 \left[x + \frac{1}{2} \right] dx$
 $= \int_{-1}^{-1/2} -1 dx + \int_{-1/2}^{1/2} 0 dx + \int_{1/2}^{3/2} 1 dx + \int_{3/2}^{5/2} 2 dx + \int_{5/2}^3 3 dx = 4$

Illustration :

Evaluate : $\int_1^4 \ln [x] dx$, $[\cdot]$ is the greatest integer function.

Sol. Let $I = \int_1^4 \ln [x] dx = \int_1^2 \ln [x] dx + \int_2^3 \ln [x] dx + \int_3^4 \ln [x] dx$
 $= 0 + \ln 2 \int_1^2 dx + \ln 3 \int_2^3 dx = \ln 2 + \ln 3 = \ln 6$

Illustration :

Find the value of following integrals

$$(a) \int_{-3}^1 |x+1| dx \quad (b) \int_0^{\pi} |\cos x - \sin x| dx$$

Sol.

$$(a) \text{ We have } |x+1| = -(x+1) \text{ for } x \leq -1; = x+1 \text{ for } x > -1$$

Hence, we have

$$\begin{aligned} \int_{-3}^1 |x+1| dx &= \int_{-3}^{-1} -(x+1) dx + \int_{-1}^1 (x+1) dx = \left[\frac{(x+1)^2}{2} \right]_{-3}^{-1} + \left[\frac{(x+1)^2}{2} \right]_{-1}^1 \\ &= -(0-2) + (2-0) = 4. \end{aligned}$$

$$(b) \text{ We have}$$

$$\begin{aligned} |\cos x - \sin x| &= \sqrt{2} \left| \sin x \left(x - \frac{\pi}{4} \right) \right| = -\sqrt{2} \sin \left(x - \frac{\pi}{4} \right) & 0 \leq x \leq \frac{\pi}{4} \\ &= \sqrt{2} \sin \left(x - \frac{\pi}{4} \right) & \frac{\pi}{4} < x \leq \pi \end{aligned}$$

Hence, we have

$$\begin{aligned} \int_0^{\pi} |\cos x - \sin x| dx &= -\sqrt{2} \int_0^{\pi/4} \sin \left(x - \frac{\pi}{4} \right) dx + \sqrt{2} \int_{\pi/4}^{\pi} \sin \left(x - \frac{\pi}{4} \right) dx \\ &= \sqrt{2} \left[\cos \left(x - \frac{\pi}{4} \right) \right]_0^{\pi/4} - \sqrt{2} \left[\cos \left(x - \frac{\pi}{4} \right) \right]_{\pi/4}^{\pi} \\ &= \sqrt{2} \left[\cos 0 - \cos \frac{\pi}{4} \right] - \sqrt{2} \left[\cos \left(x - \frac{\pi}{4} \right) - \cos 0 \right] \\ &= \sqrt{2} \left(1 - \frac{1}{\sqrt{2}} \right) - \sqrt{2} \left(\frac{1}{\sqrt{2}} - 1 \right) = \sqrt{2} - 1 + 1 + \sqrt{2} = 2\sqrt{2}. \end{aligned}$$

$$\text{P-4 : } \int_{-a}^a f(x) dx = \int_0^a (f(x) + f(-x)) dx = \begin{cases} 0 & \text{if } f(x) \text{ is odd} \\ 2 \int_0^a f(x) dx & \text{if } f(x) \text{ is even} \end{cases}$$

$$\text{Proof : } I = \int_{-a}^a f(x) dx = \int_{-a}^0 f(x) dx + \int_0^a f(x) dx \text{ Put } x = -t \text{ in first integral.}$$

$$\begin{aligned} &= \int_a^0 f(-t)(-dt) + \int_0^a f(x) dx = \int_0^a f(-t) dt + \int_0^a f(x) dx = \int_0^a f(-x) dx + \int_0^a f(x) dx \\ &= \int_0^a \{f(x) + f(-x)\} dx \end{aligned}$$

Illustration :

Show that $\int_{-1/2}^{1/2} \sec x \ln \frac{1-x}{1+x} dx = 0$.

Sol. Let $f(x) = \sec x \ln \left(\frac{1-x}{1+x} \right)$ then $f(-x) = \sec(-x) \ln \left(\frac{1+x}{1-x} \right) = -f(x)$

Illustration :

Show that $\int_{-1/2}^{1/2} \left([x] + \ln \left(\frac{1+x}{1-x} \right) \right) dx = -\frac{1}{2}$.

Sol.
$$I = \int_{-1/2}^{1/2} \left([x] + \ln \left(\frac{1+x}{1-x} \right) \right) dx = \int_{-1/2}^{1/2} [x] dx + \int_{-1/2}^{1/2} \ln \left(\frac{1+x}{1-x} \right) dx$$
$$= \int_{-1/2}^{1/2} [x] dx + 0 \quad \left(\ln \left(\frac{1+x}{1-x} \right) \text{ is an odd function} \right)$$
$$= \int_{-1/2}^0 -1 dx + \int_0^{1/2} 0 dx = -\frac{1}{2}$$

Illustration :

The value of $\int_{-1}^3 \left(\tan^{-1} \frac{x}{x^2+1} + \tan^{-1} \frac{x^2+1}{x} \right) dx$ is

- (A) π (B) 2π (C) 3π (D) $5\pi/2$

Sol.
$$I = \int_{-1}^3 \left(\tan^{-1} \frac{x}{x^2+1} + \tan^{-1} \frac{x^2+1}{x} \right) dx$$
$$= \int_{-1}^1 \left(\tan^{-1} \frac{x}{x^2+1} + \tan^{-1} \frac{x^2+1}{x} \right) dx + \int_1^3 \left(\tan^{-1} \frac{x}{x^2+1} + \tan^{-1} \frac{x^2+1}{x} \right) dx$$
$$= 0 + \int_1^3 \tan^{-1} \left(\frac{x}{x^2+1} \right) + \cot^{-1} \left(\frac{x}{x^2+1} \right) dx = 0 + \int_1^3 \frac{\pi}{2} dx = 2 \times \frac{\pi}{2} = \pi \text{ Ans.}$$

Illustration :

Evaluate : $\int_{-1}^1 x^3 \tan(x^2) dx$

Sol. Since $x^3 \tan(x^2)$, $x \in [-1, 1]$ is an odd function, hence by property (4), we have

$$\int_{-1}^1 x^3 \tan(x^2) dx = 0.$$

Practice Problem

Evaluate the following definite integral

Q.1 $\int_{-\pi/2}^0 |\sin x + \cos x| dx$

Q.2 $\int_{-\sqrt{2}}^{\sqrt{2}} \frac{2x^7 + 3x^6 - 10x^5 - 7x^3 - 12x^2 + x + 1}{x^2 + 2} dx$

Q.3 $\int_{-2}^2 \frac{x^2 - x}{\sqrt{x^2 + 4}} dx$

Q.4 $\int_0^{\sqrt{3}} \sin^{-1} \frac{2x}{1+x^2} dx$

Q.5 $\int_{-1}^1 \frac{(2x^{332} + x^{998} + 4x^{1668} \cdot \sin x^{691})}{1+x^{666}} dx$

Answer key

Q.1 $2(\sqrt{2} - 1).$

Q.2 $\frac{\pi}{2\sqrt{2}} - \frac{16\sqrt{2}}{5}$

Q.3 $4\sqrt{2} - 4 \ln(\sqrt{2} + 1)$

Q.4 $\frac{\pi\sqrt{3}}{3}$

Q.5 $\frac{\pi + 4}{666}$

P - 5 : $\int_a^b f(x) dx = \int_a^b f(a+b-x) dx$ or $\int_0^a f(x) dx = \int_0^a f(a-x) dx$

Proof: $I = \int_a^b (a+b-x) dx$ Put $a+b-x = t \Rightarrow -dx = dt$ & $I = \int_b^a f(t)(-dt)$
 $= \int_a^b f(t) dt = \int_a^b f(x) dx$

P - 6 : $\int_0^{2a} f(x) dx = \int_0^a f(x) dx + \int_0^a f(2a-x) dx \Rightarrow \begin{cases} 0 & \text{if } f(2a-x) = -f(x) \\ 2 \int_0^a f(x) dx & \text{if } f(2a-x) = f(x) \end{cases}$

Proof: $I = \int_0^{2a} f(x) dx = \int_0^a f(x) dx + \int_a^{2a} f(x) dx$ Put $x = 2a - t$ in 2nd integral
 $\Rightarrow I = \int_0^a f(x) dx + \int_a^0 f(2a-t)(-dt) = \int_0^a f(x) dx + \int_0^a f(2a-x) dx$
 $= \int_0^a \{f(x) + f(2a-x)\} dx$

Illustration :

Evaluate : $\int_{50}^{100} \frac{\ln x}{\ln x + \ln(150-x)} dx$

Sol. $I = \int_{50}^{100} \frac{\ln x}{\ln x + \ln(150-x)} dx$ using P-5 $[x \rightarrow 100 + 50 - x]$

$$I = \int_{50}^{100} \frac{\ln(150-x)}{\ln(150-x) + \ln x} dx$$

$$I + I = \int_{50}^{100} 1 dx = 50 \Rightarrow I = 25$$

Illustration :

Evaluate : $\int_0^{\pi} \frac{dx}{1+2^{\tan x}}$

Sol. $I = \int_0^{\pi} \frac{dx}{1+2^{\tan x}}$ using P-5 $[x \rightarrow \pi - x]$

$$I = \int_0^{\pi} \frac{dx}{1+2^{-\tan x}} = \int_0^{\pi} \frac{2^{\tan x}}{1+2^{\tan x}} dx$$

$$\Rightarrow I + I = \int_0^{\pi} 1 dx = \pi \text{ or } I = \frac{\pi}{2}$$

Illustration :

Evaluate : $\int_{-\pi/4}^{\pi/4} \frac{\tan^2 x}{1+e^x} dx$

Sol. $I = \int_{-\pi/4}^{\pi/4} \frac{\tan^2 x}{1+e^x} dx$ using P-5 $[x \rightarrow 0 - x]$

$$\Rightarrow I = \int_{-\pi/4}^{\pi/4} \frac{\tan^2 x}{1+e^{-x}} dx \text{ or } I + I = \int_{-\pi/4}^{\pi/4} \tan^2 x dx = 2 \int_0^{\pi/4} (\sec^2 x - 1) dx$$

$$\Rightarrow I = [\tan x - x]_0^{\pi/4} = 1 - \frac{\pi}{4}$$

Illustration :

Find the value of following integral

$$(a) \int_0^{\pi/2} \frac{1}{1+\sqrt{\tan x}} dx \quad (b) \int_2^3 \frac{\sqrt{5-x}}{\sqrt{x}+\sqrt{5-x}} dx$$

Sol.

$$(a) \int_0^{\pi/2} \frac{1}{1+\sqrt{\tan x}} dx = \int_0^{\pi/2} \frac{\sqrt{\cos x}}{\sqrt{\cos x} + \sqrt{\sin x}} dx$$

Also, we have by property P-5

$$I = \int_0^{\pi/2} \frac{\sqrt{\cos(\pi/2-x)}}{\sqrt{\cos(\pi/2-x)} + \sqrt{\sin(\pi/2-x)}} dx = \int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx$$

Adding the above integrals, we have

$$2I = \int_0^{\pi/2} \frac{\sqrt{\sin x} + \sqrt{\cos x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx = \int_0^{\pi/2} 1 dx = [x]_0^{\pi/2} = \frac{\pi}{2}$$

$$\text{i.e. } I = \frac{\pi}{4}.$$

$$(b) \text{ Let } I = \int_2^3 \frac{\sqrt{5-x}}{\sqrt{x}+\sqrt{5-x}} dx$$

Also, we have by property P-5

$$I = \int_2^3 \frac{\sqrt{5-(5-x)}}{\sqrt{5-x} + \sqrt{5-(5-x)}} dx = \int_2^3 \frac{\sqrt{x}}{\sqrt{5-x} + \sqrt{x}} dx$$

Adding the above integrals, we have

$$2I = \int_2^3 \frac{\sqrt{5-x} + \sqrt{x}}{\sqrt{5-x} + \sqrt{x}} dx = \int_2^3 1 dx = [x]_2^3 = 1$$

$$\text{i.e. } I = \frac{1}{2}.$$

Illustration :

$$\text{Evaluate : } \int_0^{\pi/2} \frac{dx}{1+\sin x}$$

$$\text{Sol. } \int_0^{\pi/2} \frac{dx}{1+\sin x} = \int_0^{\pi/2} \frac{dx}{1+\sin(\pi/2-x)} = \int_0^{\pi/2} \frac{dx}{1+\cos x}$$

$$= \frac{1}{2} \int_0^{\pi/2} \sec^2 \frac{x}{2} dx = \frac{1}{2} \left[\frac{\tan x/2}{1/2} \right]_0^{\pi/2} = 1.$$

Illustration :

$$I = \int_0^{2\pi} \sin^4 x \, dx = k \int_0^{\pi/2} \cos^4 x \, dx \quad \text{find value of } k?$$

Sol. Let $f(x) = \sin^4 x$ then $f(2\pi - x) = \sin^4(2\pi - x) = f(x)$

using P-6 $I = 2 \int_0^{\pi} \sin^4 x \, dx$, again using P-6 $I = 4 \int_0^{\pi/2} \sin^4 x \, dx$

using P-5 $I = 4 \int_0^{\pi/2} \cos^4 x \, dx$ **Ans. 4**

Illustration :

Evaluate : $I = \int_0^{2\pi} x \cdot \cos^5 x \, dx$

Sol. Using P-5 $I = \int_0^{2\pi} (2\pi - x) \cos^5 x \, dx$

$$\Rightarrow I + I = \int_0^{2\pi} (x + 2\pi - x) \cos^5 x \, dx \quad \text{or } I = \int_0^{2\pi} \pi \cos^5 x \, dx = 2\pi \int_0^{\pi} \cos^5 x \, dx \quad [\text{using P-5}]$$

$$= 2\pi \times 0 \quad [\text{using P-6 as } \cos^5(\pi - x) = -\cos^5 x]$$

Illustration :

Evaluate : $\int_0^{\pi} \frac{\pi - x}{a^2 \cos^2 x + b^2 \sin^2 x} \, dx$

Sol. Let $I = \int_0^{\pi} \frac{\pi - x}{a^2 \cos^2 x + b^2 \sin^2 x} \, dx$ using P-5

we have

$$I = \int_0^{\pi} \frac{\pi - (\pi - x)}{a^2 \cos^2(\pi - x) + b^2 \sin^2(\pi - x)} \, dx = \int_0^{\pi} \frac{x}{a^2 \cos^2 x + b^2 \sin^2 x} \, dx$$

Adding the above integrals, we have

$$2I = \int_0^{\pi} \frac{\pi}{a^2 \cos^2 x + b^2 \sin^2 x} \, dx \quad \text{i.e.} \quad I = \frac{\pi}{2} \int_0^{\pi} \frac{\sec^2 x}{a^2 + b^2 \tan^2 x} \, dx$$

Hence, we have

$$I = \pi \int_0^{\pi/2} \frac{\sec^2 x}{a^2 + b^2 \tan^2 x} dx \quad [\text{using P-6}]$$

Putting $\tan x = t$ and $\sec^2 x dx = dt$, we have

$$I = \pi \int_0^{\infty} \frac{dt}{a^2 + b^2 t^2} = \frac{\pi}{ab} \lim_{t \rightarrow \infty} \left[\tan^{-1} \left(\frac{bt}{a} \right) \right]_0^t = \frac{\pi}{ab} \cdot \frac{\pi}{2} = \frac{\pi^2}{ab}.$$

Practice Problem

Q.1 Prove that $\int_0^{\pi/4} \ln(1 + \tan x) dx = \frac{\pi}{8} \ln 2$

Q.2 $\int_{\pi/4}^{3\pi/4} \frac{x \sin x}{1 + \sin x} dx$

Q.3 $\int_0^{\pi} \frac{(ax+b)\sec x \tan x}{4 + \tan^2 x} dx \quad (a, b > 0)$

Q.4 $\int_0^{\pi} \frac{(2x+3)\sin x}{(1 + \cos^2 x)} dx$

Q.5 $\pi \int_0^{\pi} \frac{x^2 \sin 2x \cdot \sin\left(\frac{\pi}{2} \cos x\right)}{2x - \pi} dx$

Answer key

Q.2 $\pi \left[\frac{\pi}{4} - (\sqrt{2} - 1) \right]$

Q.3 $\frac{(a\pi + 2b)\pi}{3\sqrt{3}}$

Q.4 $\frac{\pi(\pi + 3)}{2}$

Q.5 8

P-7 : $\int_0^{nT} f(x) dx = n \int_0^T f(x) dx$ where $f(T+x) = f(x)$ $n \in \mathbb{I}$

Illustration :

Show that $\int_0^{1000} e^{x-[x]} dx = 1000(e-1)$

Sol. $I = \int_0^{1000} e^{x-[x]} dx = 1000 \int_0^1 e^{x-[x]} dx \quad (e^{x-[x]} \text{ has period } 1)$

$$= 1000 \int_0^1 e^x dx = 1000 [e^x]_0^1 = 1000(e-1)$$

Illustration :

Prove that : $\int_0^{n\pi+v} |\cos x| dx = (2n + 2 - \sin v)$; where $\frac{\pi}{2} < v < \pi$ & $n \in N$.

Sol.
$$I = \int_0^{n\pi+v} |\cos x| dx = \int_0^{n\pi} |\cos x| dx + \int_{n\pi}^{n\pi+v} |\cos x| dx \quad (\text{Put } x - n\pi = t \text{ in 2nd integral})$$

$$= n \int_0^{\pi} |\cos x| dx + \int_0^v |\cos(n\pi + t)| dt = 2n \int_0^{\pi/2} |\cos x| dx + \int_0^v |\cos t| dt$$

$$= 2n + \int_0^{\pi/2} \cos t dt + \int_{\pi/2}^v -\cos t dt = 2n + 1 + 1 - \sin v = 2n + 2 - \sin v$$

Illustration :

Evaluate : $\int_0^{10} \sqrt{1 - \cos \pi x} dx$

Sol. We have $\sqrt{1 - \cos \pi x} = \sqrt{2} \left| \sin \frac{\pi x}{2} \right|$

which is a periodic function, having period $T = \frac{1}{2} \times \frac{2\pi}{\pi/2} = 2$.

Hence, we have

$$\int_0^{10} \sqrt{1 - \cos \pi x} dx = \int_0^{10} \sqrt{2} \left| \sin \frac{\pi x}{2} \right| dx = 5\sqrt{2} \int_0^2 \left| \sin \frac{\pi x}{2} \right| dx$$

$$= 5\sqrt{2} \int_0^2 \sin \left(\frac{\pi x}{2} \right) dx = 5\sqrt{2} \left[\frac{-\cos(\pi x/2)}{\pi/2} \right]_0^2 = \frac{20\sqrt{2}}{\pi}.$$

(B) DERIVATIVES OF ANTIDERIVATIVES (LEIBNITZ RULE) :

If f is continuous then

$$\frac{d}{dx} \int_{g(x)}^{h(x)} f(t) dt = f(h(x)) \cdot h'(x) - f(g(x)) \cdot g'(x) \quad (\text{integral of a continuous function is always differentiable})$$

Proof: Let $\int f(t)dt = F(t) + c$ then $\int_{g(x)}^{h(x)} f(t)dt = F(h(x)) - F(g(x))$

$$\Rightarrow \frac{d}{dx} \int_{g(x)}^{h(x)} f(t)dt = F'(h(x)) h'(x) - F'(g(x)) g'(x) = f(h(x)) h'(x) - f(g(x)) g'(x)$$

Illustration :

Let $G(x) = \int_2^{x^2} \frac{dt}{1+\sqrt{t}}$ ($x > 0$). Find $G'(9)$.

Sol. $G'(x) = \frac{1}{1+\sqrt{x^2}} \cdot 2x - 0 = \frac{2x}{1+x} \Rightarrow G'(9) = \frac{2 \times 9}{1+9} = \frac{9}{5}$

Illustration :

If $f(x) = \int_{e^{2x}}^{e^{3x}} \frac{t}{\ln t} dt$ $x > 0$. Find derivative of $f(x)$ w.r.t. $\ln x$ when $x = \ln 2$.

Sol. $f'(x) = \frac{e^{3x}}{\ln(e^{3x})} \cdot 3e^{3x} - \frac{e^{2x}}{\ln(e^{2x})} \cdot 2e^{2x} = \frac{e^{6x}}{x} - \frac{e^{4x}}{x}$

$$f'(\ln 2) = \frac{e^{6 \ln 2} - e^{4 \ln 2}}{\ln 2} = \frac{2^6 - 2^4}{\ln 2} = \frac{48}{\ln 2}$$

Illustration :

Evaluate : $\lim_{x \rightarrow 0} \frac{\int_0^x (1 - \cos 2x) dx}{x \int_0^x \tan x dx}$.

Sol. $\lim_{x \rightarrow 0} \frac{\int_0^x (1 - \cos 2x) dx}{x \int_0^x \tan x dx} = \lim_{x \rightarrow 0} \frac{\int_0^x (1 - \cos 2x) dx}{x^3} \cdot \frac{x^2}{\int_0^x \tan x dx}$

Now, we have

$$\lim_{x \rightarrow 0} \frac{\int_0^x (1 - \cos 2x) dx}{x^3} \left(\frac{0}{0} \right) = \lim_{x \rightarrow 0} \frac{1 - \cos 2x}{3x^2} = \frac{2}{3}$$

and $\lim_{x \rightarrow 0} \frac{x^2}{\int_0^x \tan x dx} \left(\frac{0}{0} \right) = \lim_{x \rightarrow 0} \frac{2x}{\tan x} = 2$.

Hence, we have $L = \frac{4}{3}$.

Practice Problem

Q.1 Evaluate $\lim_{x \rightarrow 0} \frac{\int_0^{x^2} \cos t^2 dt}{x \sin x}$

Q.2 Evaluate $\int_0^{200\pi} \sqrt{1 + \cos x} dx$

Q.3 Value of $\lim_{x \rightarrow 0} \frac{\int_0^x x e^{t^2} dt}{1 - e^{x^2}}$ is

- (A) -1 (B) $-\frac{1}{2}$ (C) 0 (D) -2

Q.4 If $\int_0^{n\pi} \frac{x |\sin x|}{1 + |\cos x|} dx$ ($n \in \mathbb{N}$) is equal to $100\pi \ln 2$, then find the value of n .

Q.5 If $y = x^{\int \ln t dt}$, find $\frac{dy}{dx}$ at $x = e$.

Q.6 Let $g(x) = x^c \cdot e^{2x}$ & let $f(x) = \int_0^x e^{2t} \cdot (3t^2 + 1)^{1/2} dt$. For a certain value of 'c', the limit of $\frac{f'(x)}{g'(x)}$ as $x \rightarrow \infty$ is finite and non zero. Determine the value of 'c' and the limit.

Answer key

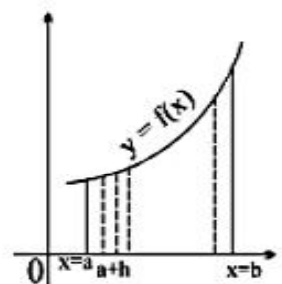
Q.1 1 Q.2 $400\sqrt{2}$ Q.3 A Q.4 10
Q.5 $1 + e$ Q.6 $c = 1$ and $\lim_{x \rightarrow \infty} \frac{f'(x)}{g'(x)}$ will be $\frac{\sqrt{3}}{2}$

(C) DEFINITE INTEGRAL AS A LIMIT OF SUM :

Fundamental theorem of integral calculus :

$$\int_a^b f(x) dx = \lim_{\substack{h \rightarrow 0 \\ n \rightarrow \infty}} h [f(a) + f(a+h) + f(a+2h) + \dots + f(a+(n-1)h)]$$

or $\int_a^b f(x) dx = \lim_{\substack{h \rightarrow 0 \\ n \rightarrow \infty}} h \sum_{r=0}^{n-1} f(a+rh)$ where $b-a = nh$



Note: Evaluating a definite integral by evaluating the limit of a sum is called evaluating definite integral by first principle or by a b initio method.

Put $a=0$ & $b=1 \Rightarrow nh=1$, we have

$$\int_0^1 f(x) dx = \frac{1}{n} \sum_{r=0}^{n-1} f\left(\frac{r}{n}\right); \text{ replace } \frac{1}{n} \rightarrow dx; \Sigma \rightarrow \int; \frac{r}{n} \rightarrow x$$

Illustration :

Evaluate $\int_a^b \cos x \, dx$ as the limit of a sum.

Sol. We have
$$\int_a^b \cos x \, dx = \lim_{h \rightarrow 0} \sum_{r=1}^n h f(a+rh) = \lim_{h \rightarrow 0} \sum_{r=1}^n h \cos(a+rh)$$

$$= \lim_{h \rightarrow 0} h [\cos(a+h) + \cos(a+2h) + \dots + \cos(a+nh)]$$

Now, let $S = \cos(a+h) + \cos(a+2h) + \dots + \cos(a+nh)$. Multiplying both sides by $2 \sin \frac{h}{2}$, we have

$$\begin{aligned} \left(2 \sin \frac{h}{2}\right) S &= 2 \sin \frac{h}{2} \cos(a+h) + 2 \sin \frac{h}{2} \cos(a+2h) + \dots + 2 \sin \frac{h}{2} \cos(a+nh) \\ &= \sin\left(a + \frac{3}{2}h\right) - \sin\left(a + \frac{1}{2}h\right) + \sin\left(a + \frac{5}{2}h\right) - \sin\left(a + \frac{3}{2}h\right) \\ &\quad + \dots + \sin\left(a + \frac{2n+1}{2}h\right) - \sin\left(a + \frac{2n-1}{2}h\right) \\ &= \sin\left(a + \frac{2n+1}{2}h\right) - \sin\left(a + \frac{1}{2}h\right) \\ &= \sin\left(a + nh + \frac{h}{2}\right) - \sin\left(a + \frac{h}{2}\right) = \sin\left(b + \frac{h}{2}\right) - \sin\left(a + \frac{h}{2}\right) \end{aligned}$$

Hence, we have

$$\int_a^b \cos x \, dx = \lim_{h \rightarrow 0} \frac{h \left[\sin\left(b + \frac{h}{2}\right) \right] - \sin\left(a + \frac{h}{2}\right)}{2 \sin\left(\frac{h}{2}\right)} = \sin b - \sin a.$$

Illustration :

Find the value of $\lim_{n \rightarrow \infty} \frac{1}{n} + \frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{4n}$

Sol.
$$S = \lim_{n \rightarrow \infty} \sum_{r=0}^{3n} \frac{1}{n+r} = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=0}^{3n} \frac{1}{1 + \left(\frac{r}{n}\right)} = \int_0^3 \frac{1}{1+x} dx = [\ln(1+x)]_0^3 = \ln 4$$

Illustration :

Evaluate $\lim_{n \rightarrow \infty} \prod_{r=1}^n \left(\frac{n+r}{n} \right)^{1/n}$.

Sol. We have

$$S = \lim_{n \rightarrow \infty} \prod_{r=1}^n \left(\frac{n+r}{n} \right)^{1/n} = \lim_{n \rightarrow \infty} \left[\frac{n+1}{n} \cdot \frac{n+2}{n} \cdot \dots \cdot \frac{n+n}{n} \right]^{1/n}$$

Taking in both sides, we have

$$\ln S = \lim_{n \rightarrow \infty} \frac{1}{n} \left[\ln \left(\frac{n+1}{n} \right) + \ln \left(\frac{n+2}{n} \right) + \dots + \ln \left(\frac{n+n}{n} \right) \right]$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \ln \left(1 + \frac{r}{n} \right) = \int_0^1 \ln(1+x) dx$$

$$= \ln 2 - (1 - \ln 2) = \ln 4 - 1 = \ln \left(\frac{4}{e} \right)$$

$$\therefore S = \frac{4}{e}.$$

Illustration :

If $n \rightarrow \infty$, then find the limit of $\frac{1}{n} \sum_{r=1}^n \sin^{2k} \left(\frac{r\pi}{2n} \right)$.

Sol. Let $P = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \sin^{2k} \left(\frac{r\pi}{2n} \right) = \int_0^1 \sin^{2k} \left(\frac{\pi}{2} x \right) dx$

Put $\frac{\pi}{2} x = t \quad \therefore dx = \frac{2}{\pi} dt$

$$= \frac{2}{\pi} \int_0^{\pi/2} \sin^{2k} t dt = \frac{2}{\pi} \frac{(2k-1)(2k-3)(2k-5) \dots 3.1}{2k(2k-2)(2k-4) \dots 4.2} \cdot \frac{\pi}{2}$$

$$= \frac{2k(2k-1)(2k-2)(2k-3) \dots 3.2.1}{[2k(2k-2)(2k-4) \dots 4.2]^2}$$

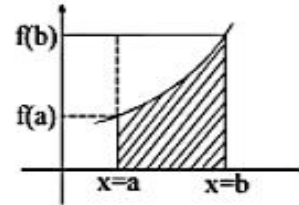
Hence $P = \frac{2k!}{(2^k k!)^2} = \frac{2k!}{2^{2k} (k!)^2}$

(D) ESTIMATION OF DEFINITE INTEGRAL AND GENERAL INEQUALITIES IN INTEGRATION:

Not all integrals can be evaluated using the technique discussed so far. In this situation we try to obtain the interval in which value of integral may lie by using following method.

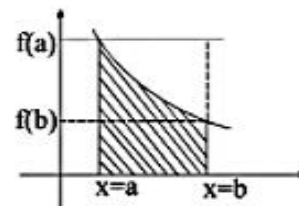
- (a) For a monotonic increasing function in (a, b)

$$(b-a)f(a) < \int_a^b f(x) dx < (b-a)f(b)$$



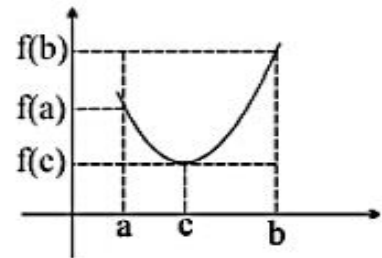
- (b) For a monotonic decreasing function in (a, b)

$$f(b) \cdot (b-a) < \int_a^b f(x) dx < (b-a)f(a)$$



- (c) For a non monotonic function in (a, b)

$$f(c) \cdot (b-a) < \int_a^b f(x) dx < (b-a)f(b)$$



- (d) In addition to this note that

$$\left| \int_a^b f(x) dx \right| \leq \int_a^b |f(x)| dx \quad \text{inequality holds when } f(x) \text{ lies completely above the } x\text{-axis}$$

- (e) If $h(x) \leq f(x) \leq g(x) \forall x \in [a, b]$ then $\int h(x) dx < \int f(x) dx < \int g(x) dx$

Illustration :

Show that $1 < \int_0^{\pi/2} \frac{\sin x}{x} dx < \frac{\pi}{2}$.

Sol. $f(x) = \frac{\sin x}{x}$ or $f'(x) = \frac{x \cos x - \sin x}{x^2} = \frac{\cos x}{x^2} [x - \tan x]$

$$\Rightarrow f'(x) < 0 \quad \text{hence } f(x)_{\min} = \frac{2}{\pi} \quad f(x)_{\max} = 1.$$

$$\Rightarrow \frac{2}{\pi} \left(\frac{\pi}{2} - 0 \right) < I < 1 \left(\frac{\pi}{2} - 0 \right) \quad \text{or } 1 < I < \frac{\pi}{2}.$$

Illustration :

Show that $\frac{1}{4} \leq \int_0^1 \frac{dx}{1+x^2+2x^5} \leq 1$.

Sol. Consider the following function $f(x) = \frac{1}{1+x^2+2x^5}, x \in [0, 1]$

In the interval $[0, 1]$, $f(x)$ is strictly decreasing, therefore we have

$$f(1) \leq f(x) \leq f(0) \quad \text{i.e.} \quad \frac{1}{4} \leq f(x) \leq 1$$

Hence, we have

$$(1-0) \frac{1}{4} \leq \int_0^1 f(x) dx \leq (1-0) 1 \quad [\text{by property (7)}]$$

$$\text{i.e.} \quad \frac{1}{4} \leq \int_0^1 f(x) dx \leq 1 \quad \text{which is the desired result.}$$

Illustration :

Prove that $\frac{\pi}{6} < \int_0^1 \frac{dx}{\sqrt{4-x^2-x^3}} < \frac{\pi\sqrt{2}}{8}$

Sol. $4-2x^2 \leq 4-x^2-x^3 \leq 4-x^2$

$$\Rightarrow \frac{1}{\sqrt{4-2x^2}} \geq \frac{1}{\sqrt{4-x^2-x^3}} \geq \frac{1}{\sqrt{4-x^2}} \Rightarrow \int_0^1 \frac{dx}{\sqrt{4-x^2}} < \int_0^1 \frac{dx}{\sqrt{4-x^2-x^3}} < \int_0^1 \frac{dx}{\sqrt{4-2x^2}}$$

$$\Rightarrow \left[\sin^{-1} \left(\frac{x}{2} \right) \right]_0^1 < I < \frac{1}{\sqrt{2}} \left[\sin^{-1} \frac{x}{\sqrt{2}} \right]_0^1 \quad \text{or} \quad \frac{\pi}{6} < I < \frac{\sqrt{2}}{8} \pi$$

Practice Problem

Q.1 $\lim_{n \rightarrow \infty} \frac{\sqrt{1} + \sqrt{2} + \sqrt{3} + \dots + 2\sqrt{n}}{n\sqrt{n}}$.

Q.2 $\lim_{n \rightarrow \infty} \frac{[(n+1)(n+2)\dots(n+n)]^{1/n}}{n}$

Q.3 Prove that $\frac{e-1}{3} < \int_1^e \frac{dx}{2+\ln x} < \frac{e-1}{2}$

Q.4 Prove the inequalities: $2e^{-1/4} < \int_0^2 e^{x^2-x} dx < 2e^2$.

Answer key

Q.1 $\frac{16}{3}$

Q.2 $\frac{4}{e}$

(E) WALLI'S THEOREM :

$$\int_0^{\pi/2} \sin^n x \cos^m x \, dx = \frac{[(n-1)(n-3)\dots 1 \text{ or } 2] [(m-1)(m-3)\dots 1 \text{ or } 2]}{(m+n)(m+n-2)\dots 1 \text{ or } 2} K$$

(m, n are non-negative integer)

$$\text{where } K = \begin{cases} \frac{\pi}{2} & \text{if } m, n \text{ both are even} \\ 1 & \text{otherwise} \end{cases}$$

Illustration :

Evaluate : $\int_0^{2\pi} x \sin^6 x \cos^4 x \, dx$

Sol. $I = \int_0^{2\pi} x \sin^6 x \cos^4 x \, dx$ using P5

$$I = \int_0^{2\pi} (2\pi - x) \sin^6 x \cos^4 x \, dx \Rightarrow I + I = \int_0^{2\pi} 2\pi \sin^6 x \cos^4 x \, dx$$

or $I = \pi \int_0^{2\pi} \sin^6 x \cos^4 x \, dx$ Using P6 twice

$$I = 4\pi \int_0^{\pi/2} \sin^6 x \cos^4 x \, dx = 4\pi \frac{(5 \cdot 3 \cdot 1)(3 \cdot 1)}{10 \cdot 8 \cdot 6 \cdot 4 \cdot 2} = \frac{3\pi^2}{128} \quad \text{Ans.}$$

Illustration :

Evaluate : $\int_0^{3\pi/2} \cos^4 3x \cdot \sin^2 6x \, dx$

Sol. $I = \int_0^{3\pi/2} \cos^4 3x \cdot \sin^2 6x \, dx = \int_0^{3\pi/2} 4 \sin^2(3x) \cos^6(3x) \, dx$

Put $3x = t$ to get

$$I = \int_0^{9\pi/2} 4 \cdot \sin^2 t \cdot \cos^6 t \cdot \frac{1}{3} dt = \frac{4}{3} \left[\int_0^{4\pi} \sin^2 t \cos^6 t \, dt + \int_{4\pi}^{9\pi/2} \sin^2 t \cos^6 t \, dt \right]$$

$$= \frac{4}{3} \left[4 \int_0^{\pi} \sin^2 t \cos^6 t \, dt + \int_{4\pi}^{9\pi/2} \sin^2 t \cos^6 t \, dt \right] \quad (\sin^2 t \cos^6 t \text{ has period } \pi)$$

$$= \frac{4}{3} \left[8 \int_0^{\pi/2} \sin^2 t \cos^6 t \, dt + \int_0^{\pi/2} \sin^2 t \cos^6 t \, dt \right] \quad (\text{using P-6 in 1st integral and } t - 4\pi = z \text{ in 2nd})$$

Illustration :

Evaluate : $\int_0^{\pi/2} \cos^7 x \, dx$

Sol. $I = \int_0^{\pi/2} \cos^7 x \, dx = \frac{6 \cdot 4 \cdot 2}{7 \cdot 5 \cdot 3 \cdot 1}$

REDUCTION METHOD :

For integration of type $\int_a^b (f(x))^n \, dx$

where 'n' is big natural number it is possible to reduce 'n' by some methods specially by parts.

Illustration :

Let $I_n = \int_0^1 (1-x^a)^n \, dx$. Find the ratio I_n/I_{n+1} .

Sol. We have $I_{n+1} = \int_0^1 (1-x^a)^{n+1} \, dx$

$$= \left[x(1-x^a)^{n+1} \right]_0^1 + (n+1)a \int_0^1 x^a (1-x^a)^n \, dx$$

[taking 1 as one function and integrating by parts]

$$= (n+1)a \int_0^1 (x^a - 1 + 1)(1-x^a)^n \, dx = (n+1)a \int_0^1 (1-x^a)^n \, dx - (n+1)a \int_0^1 (1-x^a)^{n+1} \, dx$$

$$= (n+1)a I_n - (n+1)a I_{n+1}$$

Simplifying, we have $\frac{I_n}{I_{n+1}} = 1 + \frac{1}{(n+1)a}$

Illustration :

Given $I_n = \int_0^{\pi/4} (\tan x)^n \, dx \quad (n \in \mathbb{N})$

Prove that $I_n + I_{n-1} = \frac{1}{n-1} \quad (n \geq 3)$. Hence find value of I_6 .

Sol. $I_n = \int_0^{\pi/4} (\tan x)^n \, dx, \quad I_{n-2} = \int_0^{\pi/4} (\tan x)^{n-2} \, dx$

$$I_n + I_{n-1} = \int_0^{\pi/4} (\tan x)^n + (\tan x)^{n-1} \, dx$$

Put $\tan x = t$ to get $I_n + I_{n-2} = \int_0^1 t^{n-2} dt = \frac{t^{n-1}}{n-1} \Big|_0^1 = \frac{1}{n-1}$

Also $I_2 = \int_0^{\pi/4} \tan^2 x dx = \int_0^{\pi/4} (\sec^2 x - 1) dx = [\tan x - x]_0^{\pi/4} = 1 - \frac{\pi}{4}$

Using $I_n + I_{n-2} = \frac{1}{n-1}$, $I_4 + I_2 = \frac{1}{3}$ and $I_6 + I_4 = \frac{1}{5}$

$\Rightarrow I_6 = I_2 - \frac{2}{\sqrt{5}}$ or $I_6 = 1 - \frac{\pi}{4} - \frac{2}{15} = \frac{13}{15} - \frac{\pi}{4}$

(F) SOME INTEGRALS WHICH CANNOT BE FOUND IN TERMS OF KNOWN ELEMENTARY FUNCTIONS :

- (1) $\int \frac{\sin x}{x} dx$ (2) $\int \frac{\cos x}{x} dx$ (3) $\int \sqrt{\sin x} dx$ (4) $\int \sin x^2 dx$
 (5) $\int \cos x^2 dx$
 (6) $\int x \tan x dx$ (7) $\int e^{-x^2} dx$ (8) $\int e^{x^2} dx$ (9) $\int \frac{x^3}{1+x^5} dx$
 (10) $\int (1+x^2)^{1/3} dx$ (11) $\int \frac{dx}{\ln x}$ (12) $\int \sqrt{1+k^2 \sin^2 x} dx \quad k \in \mathbb{R}$

Practice Problem

Q1 Let $U_{10} = \int_0^{\pi/2} x \sin^{10} x dx$, then find the value of $\left(\frac{100U_{10}-1}{U_8} \right)$.

Q.2 If $U_n = \int_0^{\pi/2} \frac{\sin^2 nx}{\sin^2 x} dx$, then show that $U_1, U_2, U_3, \dots, U_n$ constitute an AP.
 Hence or otherwise find the value of U_n .

Q.3 Evaluate: $\int_0^{4\pi} \sin^4 x \cos^2 x dx$

Answer key

Q1 90

Q.2 $U_n = \frac{n\pi}{2}$

Q.3 $\frac{\pi}{4}$

Solved Examples

Q.1 Prove that $\int_0^{\infty} \frac{dx}{\left[x + \sqrt{1+x^2}\right]^2} = \frac{n}{n^2-1} \quad (n > 1)$

Sol. L.H.S. = $\int_0^{\infty} \frac{dx}{\left(x + \sqrt{1+x^2}\right)^n}$ Put $x + \sqrt{1+x^2} = t$ (i)

$\therefore \left(1 + \frac{x}{\sqrt{1+x^2}}\right) dx = dt$ when $x \rightarrow 0$ then $t \rightarrow 1$ $x \rightarrow \infty$ then $t \rightarrow \infty$

$\Rightarrow \frac{t dx}{\sqrt{1+x^2}} = dt \Rightarrow dx = \frac{\sqrt{1+x^2}}{t} dt \Rightarrow \sqrt{1+x^2} - x = \frac{1}{t}$ (ii)

Adding (i) & (ii) we get $2\sqrt{1+x^2} = t + \frac{1}{t}$

$\Rightarrow \sqrt{1+x^2} = \frac{t^2+1}{2t} \therefore dx = \frac{(t^2+1)}{2t^2} dt$

Hence L.H.S. = $\int_1^{\infty} \frac{(t^2+1)dt}{2t^2t^n} = \frac{1}{2} \int_1^{\infty} \left(\frac{1}{t^n} + \frac{1}{t^{n+2}}\right) dt = \frac{1}{2} \left[-\frac{1}{(n-1)t^{n-1}} - \frac{1}{(n+1)t^{n+1}} \right]_1^{\infty}$
 $= \frac{1}{2} \left[(-0-0) - \left(-\frac{1}{n-1} - \frac{1}{n+1}\right) \right] = \frac{1}{2} \left[\frac{1}{n-1} + \frac{1}{n+1} \right] \quad (\because n > 1)$
 $= \frac{n}{n^2-1} = \text{R.H.S.}$

Q.2 If $f(x) = \int_0^x \frac{e^t}{t} dt, x > 0$. Prove that $\int_1^x \frac{e^t}{(t+a)} = e^{-a} [f(x+a) - f(1+a)]$.

Sol. R.H.S. = $e^{-a} [f(x+a) - f(1+a)] = e^{-a} \left[\int_0^{x+a} \frac{e^t}{t} dt - \int_0^{1+a} \frac{e^t}{t} dt \right] = e^{-a} \left[\int_0^{x+a} \frac{e^t}{t} dt + \int_{1+a}^0 \frac{e^t}{t} dt \right]$
 $= e^{-a} \int_{1+a}^{x+a} \frac{e^t}{t} dt$ Put $t = a + y \therefore dt = dy$
 $= e^{-a} \int_1^x \frac{e^{a+y}}{(a+y)} dy = e^{-a} \cdot e^a \int_1^x \frac{e^y}{(a+y)} dy = \int_1^x \frac{e^y}{a+y} dy = \int_1^x \frac{e^t}{(a+t)} dt \quad (\text{by Prop.})$
 $= \text{L.H.S.}$

Q.3 Evaluate $\int_{-\pi}^{\pi} |x \sin[x^2 - \pi]| dx$, $[\cdot]$ is the greatest integer function.

Sol. Let $I = \int_{-\pi}^{\pi} |x \sin[x^2 - \pi]| dx = 2 \int_0^{\pi} |x \sin[x^2 - \pi]| dx$ [it is even function]

$$\begin{aligned}
 I &= 2 \left[\int_0^{\sqrt{\pi-3}} |x \sin[x^2 - \pi]| dx + \int_{\sqrt{\pi-3}}^{\sqrt{\pi-2}} |x \sin[x^2 - \pi]| dx + \int_{\sqrt{\pi-2}}^{\sqrt{\pi-1}} |x \sin[x^2 - \pi]| dx \right. \\
 &\quad \left. + \int_{\sqrt{\pi-1}}^{\sqrt{\pi}} |x \sin[x^2 - \pi]| dx + \dots + \int_{\sqrt{\pi+6}}^{\pi} |x \sin[x^2 - \pi]| dx \right] \\
 &= 2 \left[\int_0^{\sqrt{\pi-3}} x \sin 4 dx + \int_{\sqrt{\pi-3}}^{\sqrt{\pi-2}} x \sin 3 dx + \int_{\sqrt{\pi-2}}^{\sqrt{\pi-1}} x \sin 2 dx + \int_{\sqrt{\pi-1}}^{\sqrt{\pi}} x \sin 1 dx + 0 \right. \\
 &\quad \left. + \int_{\sqrt{\pi+2}}^{\sqrt{\pi+1}} x \sin 1 dx + 0 + \int_{\sqrt{\pi+1}}^{\sqrt{\pi+2}} x \sin 1 dx + \dots + \int_{\sqrt{\pi+6}}^{\pi} x \sin 6 dx \right] \\
 &= \{ \sin 4 (\pi - 3) + \sin 3(1) + \sin 2(1) + \sin 1(1) + \dots + \sin 1(1) + \dots \\
 &\quad + \sin 1(1) + \dots + \sin 6(\pi^2 - \pi - 6) \} \\
 &= 2 \sin 1 + 2 \sin 2 + 3 \sin 3 + (\pi - 2) \sin 4 + \sin 5 + (\pi^2 - \pi - 6) \sin 6
 \end{aligned}$$

Q.4 Show that $\frac{\pi}{3\sqrt{3}} \leq \int_0^1 \frac{dx}{1+x^2+2x^5} \leq \frac{\pi}{4}$.

Sol. We have $1+x^2+2x^5 \geq 1+x^2$
 and $1+x^2+2x^5 \leq 1+x^2+2x^2 = 1+3x^2$ [$x^5 < x^2$ on $[0, 1]$]

$$\text{Hence, we have } \frac{1}{1+3x^2} \leq \frac{1}{1+x^2+2x^5} \leq \frac{1}{1+x^2}$$

$$\text{i.e. } \int_0^1 \frac{dx}{1+3x^2} \leq \int_0^1 \frac{dx}{1+x^2+2x^5} \leq \int_0^1 \frac{dx}{1+x^2} \quad [\text{by property (8)}]$$

$$\text{i.e. } \left[\frac{\tan^{-1} \sqrt{3} x}{\sqrt{3}} \right]_0^1 \leq \int_0^1 \frac{dx}{1+x^2+2x^5} \leq [\tan^{-1} x]_0^1 \quad \text{i.e. } \frac{\pi}{3\sqrt{3}} \leq \int_0^1 \frac{dx}{1+x^2+2x^5} \leq \frac{\pi}{4}$$

which is the desired result.

Q.5 Evaluate $\int_{-1}^1 \frac{|\sin x|}{\sin x} dx$

Sol. We have $\frac{|\sin x|}{\sin x} = -1, -1 \leq x \leq 0 = 1, 0 \leq x \leq 1 \Rightarrow \frac{|\sin x|}{\sin x}, x \in [-1, 1]$ is an odd function

$$\text{Hence, by property (4), we have } \int_{-1}^1 \frac{|\sin x|}{\sin x} = 0.$$

Q.6 Evaluate: $\int_0^2 [x^2 - 1] dx$ where $[x]$ represents integral part of x .

Sol. We have $[x^2 - 1] = -1, 0 \leq x < 1 = 0, 1 \leq x < \sqrt{2} = 1, \sqrt{2} \leq x < \sqrt{3} = 2, \sqrt{3} \leq x < 2$.

$$\begin{aligned} \text{Hence, we have } \int_0^2 [x^2 - 1] dx &= \int_0^1 -1 dx + \int_1^{\sqrt{2}} 0 dx + \int_{\sqrt{2}}^{\sqrt{3}} 1 dx + \int_{\sqrt{3}}^2 1 dx \\ &= [-x]_0^1 + 0 + [x]_{\sqrt{2}}^{\sqrt{3}} + [x]_{\sqrt{3}}^2 = -1 + \sqrt{3} - \sqrt{2} + 2 - \sqrt{3} = 1 - \sqrt{2}. \end{aligned}$$

Q.7 Evaluate: $\int_1^3 \frac{dx}{\sqrt{x+1} - \sqrt{x-1}}$

$$\begin{aligned} \text{Sol. We have } \int_1^3 \frac{dx}{\sqrt{x+1} - \sqrt{x-1}} &= \int_1^3 \frac{\sqrt{x+1} + \sqrt{x-1}}{2} dx = \frac{1}{2} \left[\frac{(x+1)^{3/2}}{3/2} + \frac{(x-1)^{3/2}}{3/2} \right]_1^3 \\ &= \frac{1}{3} [(x+1)^{3/2} + (x-1)^{3/2}]_1^3 = \frac{1}{3} [4^{3/2} + 2^{3/2} - 2^{3/2}] = \frac{8}{3}. \end{aligned}$$

Q.8 Evaluate the definite integral $\int_0^{\pi/2} \ln(\tan x + \cot x) dx$

$$\begin{aligned} \text{Sol. Let } I &= \int_0^{\pi/2} \ln(\tan x + \cot x) dx = \int_0^{\pi/2} \ln\left(\frac{\sin x}{\cos x} + \frac{\cos x}{\sin x}\right) dx = \int_0^{\pi/2} \ln\left(\frac{1}{\sin x \cos x}\right) dx \\ &= \int_0^{\pi/2} \ln\left(\frac{2}{\sin 2x}\right) dx = \int_0^{\pi/2} \ln 2 dx - \int_0^{\pi/2} \ln(\sin 2x) dx = \frac{\pi}{2} \ln 2 - \int_0^{\pi/2} \ln(\sin 2x) dx. \end{aligned}$$

Let us put $2x = y$ and $2 dx = dy$ in the second integral on the RHS. Also, when $x = 0$, then $y = 0$ and

$$\text{when } x = \frac{\pi}{2}, \text{ then } y = \pi. \text{ Hence, we have } \int_0^{\pi/2} \ln(\sin 2x) dx = \frac{1}{2} \int_0^{\pi} \ln(\sin y) dy$$

$$= \frac{1}{2} \cdot 2 \int_0^{\pi/2} \ln(\sin y) dy \quad [\ln(\sin y) = \ln \sin \pi - y]$$

$$= -\frac{\pi}{2} \ln 2$$

$$\text{Hence, we have } I = \frac{\pi}{2} \ln 2 + \frac{\pi}{2} \ln 2 = \pi \ln 2.$$

Q.9 Find $f(x)$ if it satisfies the relation $f(x) = e^x + \int_0^1 (x + ye^x) f(y) dy$.

Sol. We have $f(x) = e^x + x \int_0^1 f(y) dy + e^x \int_0^1 y f(y) dy$

$$= e^x \left(1 + \int_0^1 y f(y) dy \right) + x \int_0^1 f(y) dy = ae^x + bx \text{ (say)}$$

where a, b are constants, given by $a = 1 + \int_0^1 y f(y) dy = 1 + \int_0^1 y (ae^y + by) dy$

$$= 1 + \left[(y-1)e^y \right]_0^1 + \left[\frac{by^3}{3} \right]_0^1 = 1 + a + \frac{b}{3}$$

and $b = \int_0^1 f(y) dy = \int_0^1 (ae^y + by) dy = \left[ae^y + \frac{by^2}{2} \right]_0^1 = a(e-1) + \frac{b}{2}$

Solving, we have $b = -2$ and $a = \frac{-3}{2(e-1)}$

Hence, we have $f(x) = \frac{-3e^x}{2(e-1)} - 3x$.

Q.10 If $b = \int_0^1 \frac{e^t}{t+1} dt$, then show that $\int_{a-1}^a \frac{e^{-t}}{t-a-1} dt = be^{-a}$.

Sol. We have $I = \int_{a-1}^a \frac{e^{-t}}{t-a-1} dt = \int_{a-1}^{-a} \frac{e^y}{-y-a-1} (-dy)$ [putting $t = -y$]

$$= e^{-a} \int_{a-1}^{-a} \frac{e^{y+a}}{y+a+1} dy = e^{-a} \int_{a-1}^{-a} \frac{e^u}{u+1} du$$
 [putting $y+a=u$]
$$= -e^{-a} \int_0^1 \frac{e^u}{u+1} dx = -be^{-a} \quad \left[\int_0^1 \frac{e^t}{t+1} dt = b \text{ given} \right]$$

Q.11 Evaluate the definite integral $\int_0^1 \frac{1-x^2}{1+x^2} \cdot \frac{dx}{\sqrt{1+x^4}}$

Sol. We have $I = \int_0^1 \frac{1-x^2}{1+x^2} \cdot \frac{dx}{\sqrt{1+x^4}} = \int_0^1 \frac{\frac{1}{x^2} - 1}{x + \frac{1}{x}} \cdot \frac{dx}{\sqrt{x^2 + \frac{1}{x^2}}} = \int_0^1 \frac{-d\left(x + \frac{1}{x}\right)}{\left(x + \frac{1}{x}\right) \sqrt{\left(x + \frac{1}{x}\right)^2 - 2}}$

$$\begin{aligned}
 &= \int_{\infty}^2 \frac{-dt}{t\sqrt{t^2-2}} \quad [\text{Putting } x + \frac{1}{x} = t] \quad = \int_2^{\infty} \frac{t dt}{t^2\sqrt{t^2-2}} \quad [\text{Putting } t^2-2 = u^2] \\
 &= \int_{\sqrt{2}}^{\infty} \frac{u du}{u(u^2+2)} = \frac{1}{\sqrt{2}} \left[\tan^{-1} \frac{u}{\sqrt{2}} \right]_{\sqrt{2}}^{\infty} = \frac{1}{\sqrt{2}} \left(\frac{\pi}{2} - \frac{\pi}{4} \right) = \frac{\pi}{4\sqrt{2}}
 \end{aligned}$$

Q.12 If $I_1 = \int_0^{n\pi} f(\sin^4 x) dx$ and $I_2 = \int_0^{\pi} f(\sin^4 x) dx$. Find the value of $\frac{I_1}{I_2}$.

Sol. We have

$$I = \frac{\int_0^{n\pi} f(\sin^4 x) dx}{\int_0^{\pi} f(\sin^4 x) dx} = \frac{\int_0^{2n\left(\frac{\pi}{2}\right)} f(\sin^4 x) dx}{\int_0^{2\left(\frac{\pi}{2}\right)} f(\sin^4 x) dx} = \frac{2n \int_0^{\pi/2} f(\sin^4 x) dx}{2 \int_0^{\pi/2} f(\sin^4 x) dx} \quad [\text{period of } \sin^4 x \text{ is } \frac{\pi}{2}] = n.$$

Q.13 Prove that $\int_{1/2}^2 (\ln x)^2 dx < \int_{1/2}^2 |\ln x| dx$

Sol. In the interval $\left[\frac{1}{4}, \frac{1}{2}\right]$, $|\ln x|$ is a fraction. hence, we have $(\ln x)^2 < |\ln x|$

$$\text{i.e. } \int_{1/2}^2 (\ln x)^2 dx < \int_{1/2}^2 |\ln x| dx.$$

Q.14 Show that $\int_0^{k\pi} \sin\left[\frac{2x}{\pi}\right] dx = \frac{\pi}{2} \cdot \frac{\sin k \sin(k+1/2)}{\sin(1/2)}$.

$$\begin{aligned}
 \text{Sol. We have } I &= \int_0^{k\pi} \sin\left[\frac{2x}{\pi}\right] dx = \int_0^{\pi/2} \sin 0 dx + \int_{\pi/2}^{2\pi/2} \sin 1 dx + \int_{2\pi/2}^{3\pi/2} \sin 2 dx + \dots + \int_{2\pi/2}^{2k\pi/2} \sin(2k-1) dx \\
 &= \frac{\pi}{2} [\sin 1 + \sin 2 + \sin 3 + \dots + \sin(2k-1)] \\
 &= \frac{\pi}{2} \left[\sin \frac{1}{2} \sin 1 + \sin \frac{1}{2} \sin 2 + \sin \frac{1}{2} \sin 3 + \dots + \sin \frac{1}{2} \sin(2k-1) \right] \\
 &= \frac{\pi}{2} \frac{\sin \frac{1}{2}}{\sin \frac{1}{2}}
 \end{aligned}$$

$$= \frac{\left[\cos \frac{1}{2} - \cos \frac{3}{2} + \cos \frac{3}{2} - \cos \frac{5}{2} + \dots + \cos \left(2k - \frac{3}{2} \right) - \cos \left(2k + \frac{1}{2} \right) \right]}{2 \sin \frac{1}{2}}$$

$$= \frac{\pi}{2} \cdot \frac{\cos \frac{1}{2} - \cos \left(2k + \frac{1}{2} \right)}{2 \sin \frac{1}{2}} = \frac{\pi}{2} \cdot \frac{\sin k \sin(k + 1/2)}{\sin(1/2)}$$

Q.15 Let $I_n = \int_0^1 x^n \tan^{-1} x \, dx$. Show that $(n+1)I_{n-1} + (n-1)I_{n-2} = \frac{\pi}{2} - \frac{1}{n}$.

Sol. We have $I_n = \int_0^1 x^n \tan^{-1} x \, dx = \int_0^{\pi/4} \theta (\tan \theta)^n \sec^2 \theta \, d\theta$ [Putting $x = \tan \theta$]

$$= \left[\frac{\theta (\tan \theta)^{n+1}}{n+1} \right]_0^{\pi/4} - \int_0^{\pi/4} \frac{(\tan \theta)^{n+1}}{n+1} d\theta = \frac{\pi/4}{n+1} - \int_0^{\pi/4} \frac{(\tan \theta)^{n-1} (\sec^2 \theta - 1)}{n+1} d\theta$$

$$= \frac{\pi/4}{n+1} - \frac{1}{n+1} \int_0^{\pi/4} (\tan \theta)^{n-1} \sec^2 \theta \, d\theta + \frac{1}{n+1} \int_0^{\pi/4} (\tan \theta)^{n-1} d\theta$$

$$= \frac{\pi/4}{n+1} - \frac{1}{n+1} \left[\frac{(\tan \theta)^n}{n} \right]_0^{\pi/4} + \left(\frac{1}{n+1} \right) I$$

$$= \frac{\pi/4}{n+1} - \frac{1}{n(n+1)} + \left(\frac{1}{n+1} \right) I \quad \text{and} \quad I_{n-2} = \int_0^{\pi/4} \theta (\tan \theta)^{n-2} \sec^2 \theta \, d\theta$$

$$= \left[\frac{\theta (\tan \theta)^{n-1}}{n-1} \right]_0^{\pi/4} - \int_0^{\pi/4} \frac{(\tan \theta)^{n-1}}{n-1} d\theta = \frac{\pi/4}{n-1} \left(\frac{1}{n-1} \right) I$$

Eliminating I from equations (1) and (2), we have

$$(n+1)I_{n-1} + (n-1)I_{n-2} = \frac{\pi}{2} - \frac{1}{n} \quad \text{which is the desired result.}$$

Q.16 Evaluate : $\lim_{n \rightarrow \infty} \prod_{r=1}^n \frac{(n^2 + r^2)^{1/n}}{n^2}$

Sol. We have $S = \lim_{n \rightarrow \infty} \prod_{r=1}^n \frac{(n^2 + r^2)^{1/n}}{n^2}$

$$= \lim_{n \rightarrow \infty} \left[\left(\frac{n^2 + 1^2}{n^2} \right) \left(\frac{n^2 + 2^2}{n^2} \right) \dots \left(\frac{n^2 + n^2}{n^2} \right) \right]^{1/n}$$

Taking in both sides, we have

$$\begin{aligned}
 \ln S &= \lim_{n \rightarrow \infty} \frac{1}{n} \left[\ln \left(1 + \frac{1^2}{n^2} \right) + \ln \left(1 + \frac{2^2}{n^2} \right) + \dots + \ln \left(1 + \frac{n^2}{n^2} \right) \right] \\
 &= \lim_{n \rightarrow \infty} \frac{1}{n} \ln \left(1 + \frac{r^2}{n^2} \right) = \int_0^1 \ln(1+x^2) dx = \left[x \ln(1+x^2) \right]_0^1 - \int_0^1 x \left(\frac{2x}{1+x^2} \right) dx \\
 &= \ln 2 - \int_0^1 2 \left(1 - \frac{1}{1+x^2} \right) dx = \ln 2 - 2 \left[x - \tan^{-1} x \right]_0^1 \\
 &= \ln 2 - 2 \left(1 - \frac{\pi}{4} \right) = \ln 2 + \frac{\pi-4}{2} \quad \text{gives } S = 2e^{\left(\frac{\pi-4}{2} \right)}.
 \end{aligned}$$

Q.17 Evaluate the definite integrals $\int_0^{\pi/2} \sin x \ln(\cos x) dx$

Sol. We have

$$\begin{aligned}
 I &= \int_0^{\pi/2} \sin x \ln(\cos x) dx = \left[-\cos x \ln(\cos x) \right]_0^{\pi/2} + \int_0^{\pi/2} \cos x \cdot \frac{-\sin x}{\cos x} dx \\
 &= \lim_{x \rightarrow \pi/2} \frac{-\ln(\cos x)}{\sec x} + \left[\cos x \right]_0^{\pi/2} = \lim_{x \rightarrow \pi/2} \frac{\tan x}{\sec x \tan x} - 1 \\
 &= \lim_{x \rightarrow \pi/2} \cos x - 1 = -1.
 \end{aligned}$$

Q.18 Evaluate the following limits, using definite integral :

$$\lim_{n \rightarrow \infty} \left[\frac{1}{\sqrt{n^2}} + \frac{1}{\sqrt{n^2-1^2}} + \frac{1}{\sqrt{n^2-2^2}} + \dots + \frac{1}{\sqrt{n^2-(n-1)^2}} \right]$$

Sol. We have

$$\begin{aligned}
 &\lim_{n \rightarrow \infty} \left[\frac{1}{\sqrt{n^2}} + \frac{1}{\sqrt{n^2-1^2}} + \frac{1}{\sqrt{n^2-2^2}} + \dots + \frac{1}{\sqrt{n^2-(n-1)^2}} \right] \\
 &= \lim_{n \rightarrow \infty} \sum_{r=0}^n \frac{1}{\sqrt{n^2-r^2}} = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \frac{1}{\sqrt{1-(r/n)^2}} \quad [\text{Omitting one term will not affect the limit}] \\
 &= \int_0^1 \frac{dx}{\sqrt{1-x^2}} = \left[\sin^{-1} x \right]_0^1 = \frac{\pi}{2}.
 \end{aligned}$$

Q.19 Prove the following results, using definite integral :

$$\lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{\sqrt{n}}{\sqrt{r}(a\sqrt{n} - b\sqrt{r})^2} = \frac{2}{a(a-b)}$$

Sol. We have

$$\begin{aligned} \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{\sqrt{n}}{\sqrt{r}(a\sqrt{n} - b\sqrt{r})^2} &= \frac{2}{a(a-b)} \\ &= \int_0^1 \frac{dx}{\sqrt{x}(a - b\sqrt{x})^2} = \frac{-2}{b} \int_a^{a-b} \frac{dt}{t^2} \quad [\text{Putting } a - b\sqrt{x} = t] \\ &= \frac{2}{b} \left[\frac{1}{t} \right]_a^{a-b} = \frac{2}{b} \left(\frac{1}{a-b} - \frac{1}{a} \right) = \frac{2}{a(a-b)} \end{aligned}$$

Q.20 Evaluate : $\lim_{n \rightarrow \infty} \left[\frac{\sqrt{n}}{\sqrt{n^3}} + \frac{\sqrt{n}}{\sqrt{(n+2)^3}} + \frac{\sqrt{n}}{\sqrt{(n+4)^3}} + \frac{\sqrt{n}}{\sqrt{(n+8)^3}} + \dots \right]$

Sol. We have

$$\begin{aligned} &\lim_{n \rightarrow \infty} \left[\frac{\sqrt{n}}{\sqrt{n^3}} + \frac{\sqrt{n}}{\sqrt{(n+2)^3}} + \frac{\sqrt{n}}{\sqrt{(n+4)^3}} + \frac{\sqrt{n}}{\sqrt{(n+8)^3}} + \dots \right] \\ &= \lim_{n \rightarrow \infty} \sum_{r=0}^n \frac{\sqrt{n}}{\sqrt{(n+2r)^3}} = \lim_{n \rightarrow \infty} \frac{\sqrt{n}}{\sqrt{(n+2r)^3}} \quad [\text{omitting one term will not affect the limit}] \\ &= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \frac{n^{3/2}}{(n+2r)^{3/2}} = \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{\left\{ 1 + 2\left(\frac{r}{n}\right) \right\}^{3/2}} \\ &= \int_0^1 \frac{1}{(1+2x)^{3/2}} dx = \left[\frac{(1+2x)^{-1/2}}{-1/2} \cdot \frac{1}{2} \right]_0^1 = \left[\frac{1}{\sqrt{1+2x}} \right]_0^1 = 1 - \frac{1}{\sqrt{3}}. \end{aligned}$$